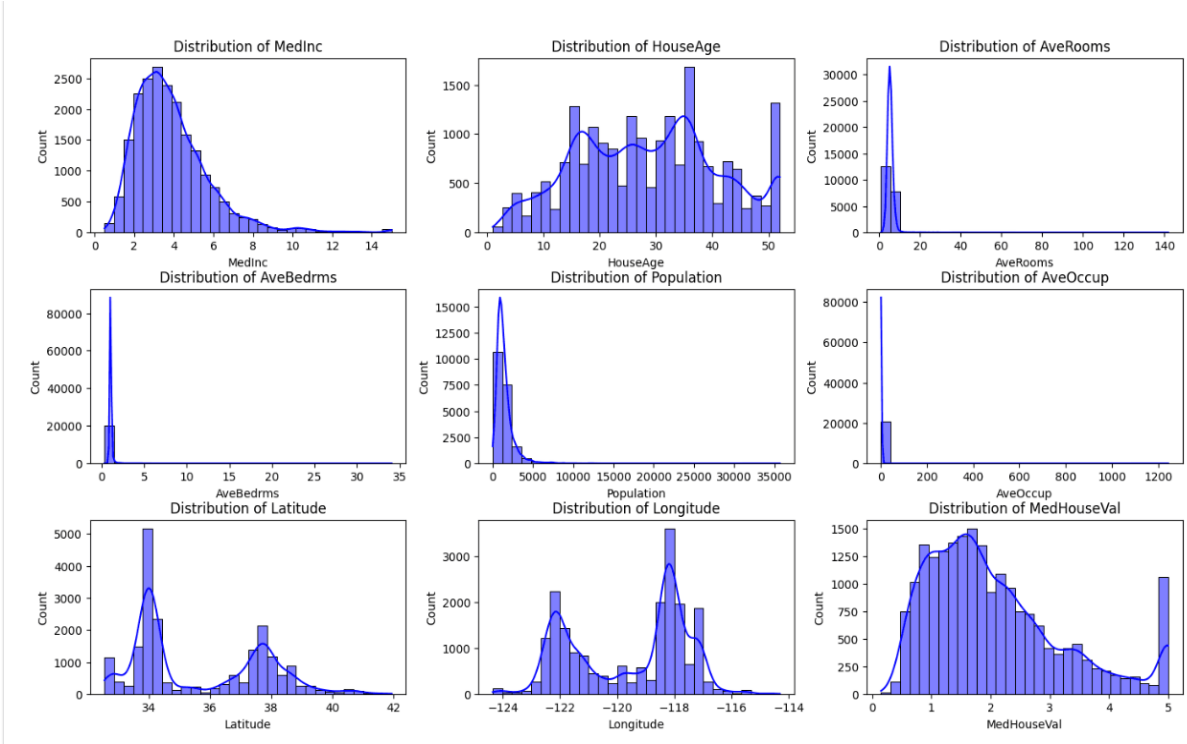
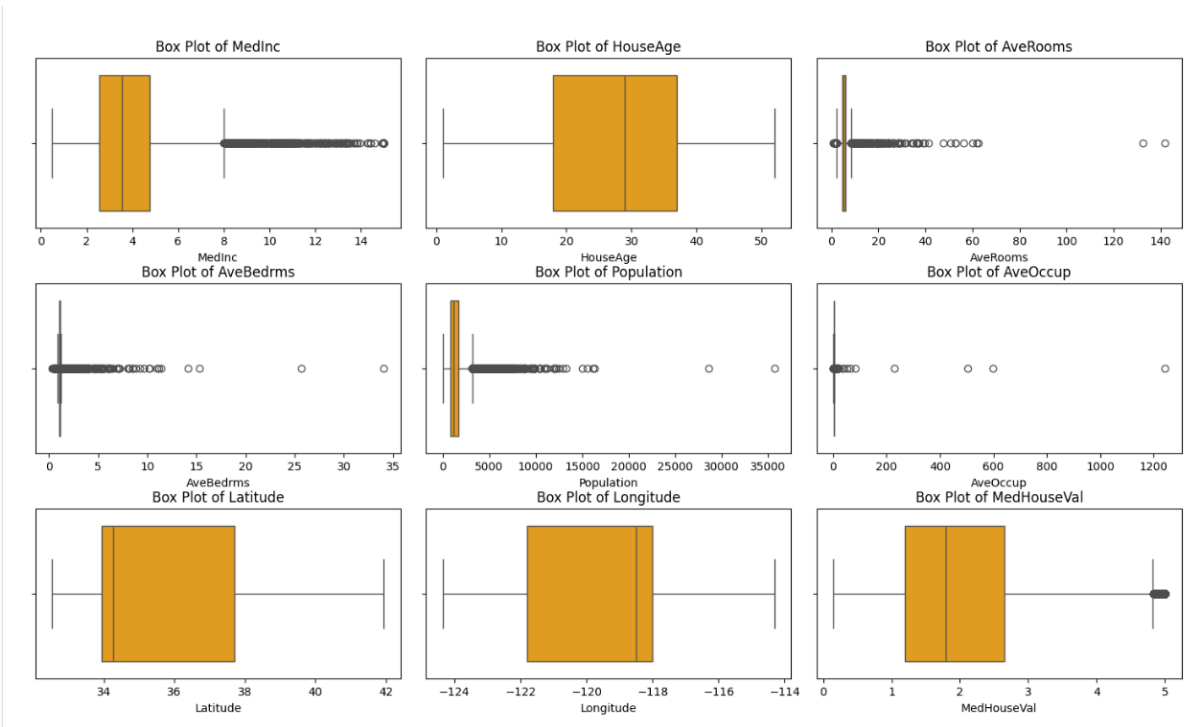
Lab 1






```
Outliers Detection:
MedInc: 681 outliers
HouseAge: 0 outliers
AveRooms: 511 outliers
AveBedrms: 1424 outliers
Population: 1196 outliers
AveOccup: 711 outliers
Latitude: 0 outliers
Longitude: 0 outliers
MedHouseVal: 1071 outliers
```

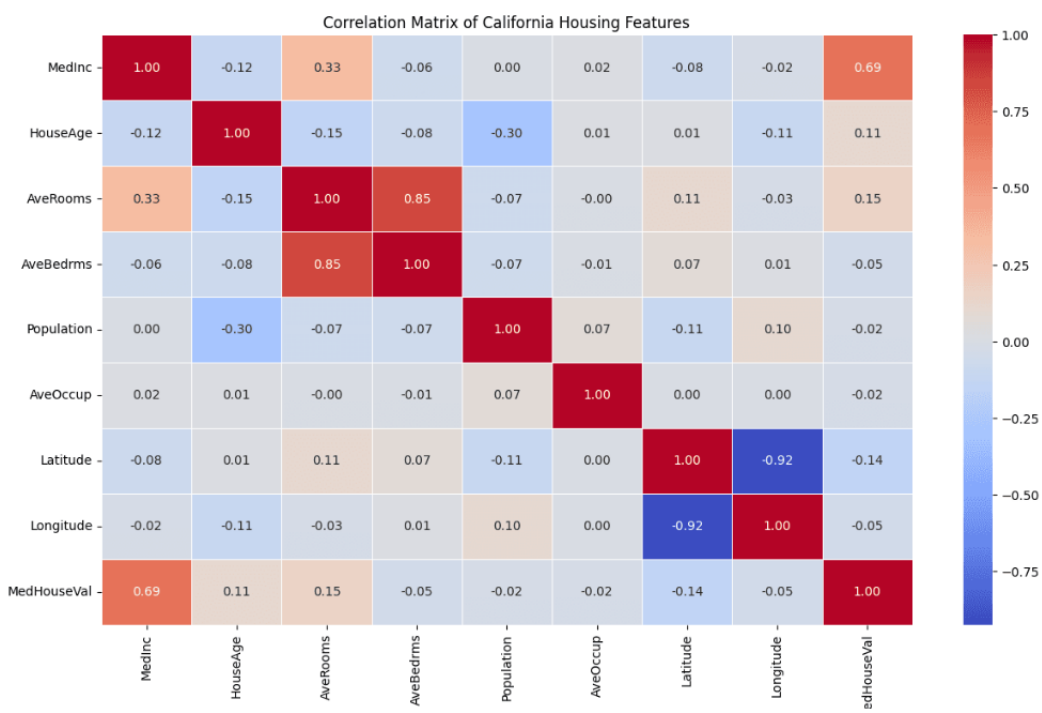
Lab1

Dataset Summary:

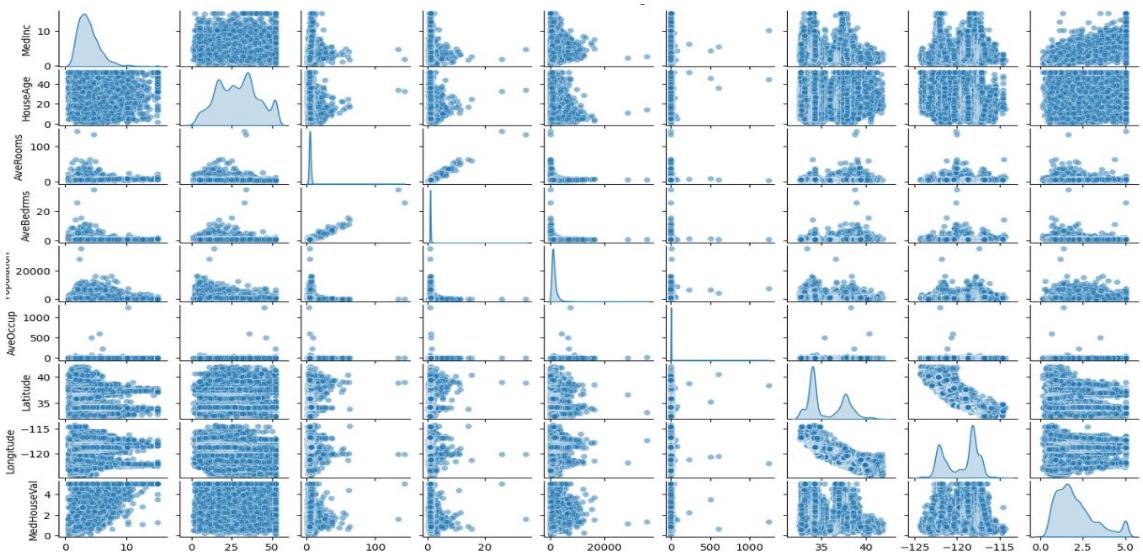
	MedInc	HouseAge	AveRooms	AveBedrms	Population \
count	20640.000000	20640.000000	20640.000000	20640.000000	20640.000000
mean	3.870671	28.639486	5.429000	1.096675	1425.476744
std	1.899822	12.585558	2.474173	0.473911	1132.462122
min	0.499900	1.000000	0.846154	0.333333	3.000000
25%	2.563400	18.000000	4.440716	1.006079	787.000000
50%	3.534800	29.000000	5.229129	1.048780	1166.000000
75%	4.743250	37.000000	6.052381	1.099526	1725.000000
max	15.000100	52.000000	141.909091	34.066667	35682.000000

	AveOccup	Latitude	Longitude	MedHouseVal
count	20640.000000	20640.000000	20640.000000	20640.000000
mean	3.070655	35.631861	-119.569704	2.068558
std	10.386050	2.135952	2.003532	1.153956
min	0.692308	32.540000	-124.350000	0.149990
25%	2.429741	33.930000	-121.800000	1.196000
50%	2.818116	34.260000	-118.490000	1.797000
75%	3.282261	37.710000	-118.010000	2.647250
max	1243.333333	41.950000	-114.310000	5.000010

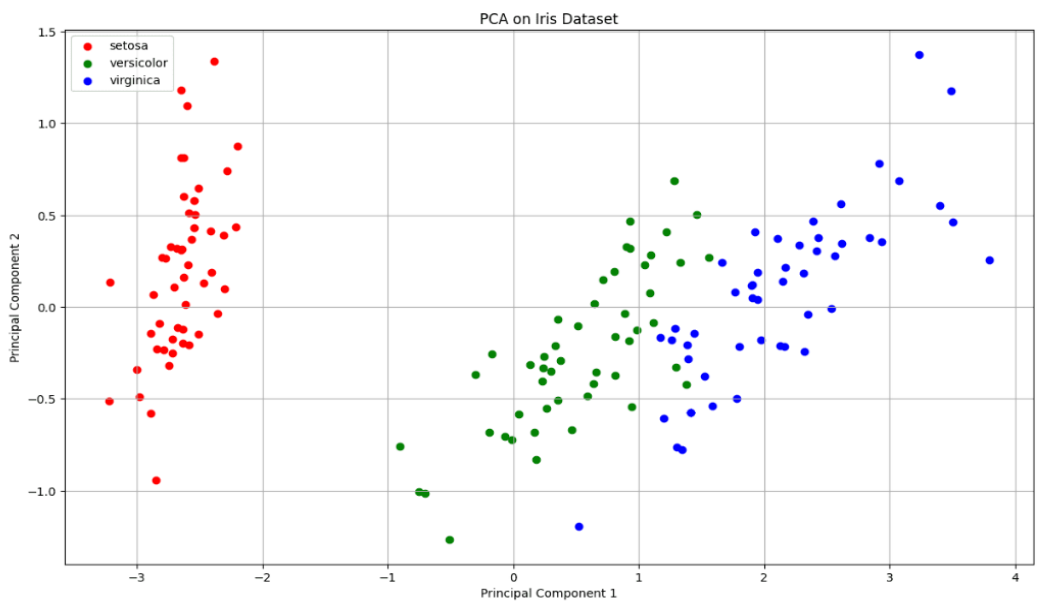
Lab 2



LAB2



LAB 3



LAB 4

output:

Training data:Outlook Temperature Humidity Windy PlayTennis

0	Sunny	Hot	High	False	No
1	Sunny	Hot	High	True	No
2	Overcast	Hot	High	False	Yes
3	Rain	Cold	High	False	Yes
4	Rain	Cold	High	True	No
5	Overcast	Hot	High	True	Yes
6	Sunny	Hot	High	False	No

The final hypothesis is: ['Overcast', 'Hot', 'High', '?']

LAB 5

--- k-Nearest Neighbors Classification ---
Training dataset: First 50 points labeled based on the rule (x <= 0.5 -> Class1, x > 0.5 -> Class2)
Testing dataset: Remaining 50 points to be classified

Results for k = 1:
Point x51 (value: 0.7565) is classified as Class2
Point x52 (value: 0.0936) is classified as Class1
Point x53 (value: 0.9394) is classified as Class2
Point x54 (value: 0.0954) is classified as Class1
Point x55 (value: 0.0315) is classified as Class1
Point x56 (value: 0.0228) is classified as Class1
Point x57 (value: 0.3583) is classified as Class1
Point x58 (value: 0.6590) is classified as Class2
Point x59 (value: 0.6483) is classified as Class2
Point x60 (value: 0.4174) is classified as Class1
Point x61 (value: 0.1488) is classified as Class1
Point x62 (value: 0.4124) is classified as Class1
Point x63 (value: 0.4557) is classified as Class1
Point x64 (value: 0.2345) is classified as Class1
Point x65 (value: 0.9202) is classified as Class2
Point x66 (value: 0.1639) is classified as Class1
Point x67 (value: 0.2964) is classified as Class1
Point x68 (value: 0.1987) is classified as Class1
Point x69 (value: 0.6599) is classified as Class2
Point x70 (value: 0.7217) is classified as Class2
Point x71 (value: 0.5584) is classified as Class2
Point x72 (value: 0.8772) is classified as Class2
Point x73 (value: 0.7663) is classified as Class2
Point x74 (value: 0.1449) is classified as Class1
Point x75 (value: 0.8393) is classified as Class2
Point x76 (value: 0.9905) is classified as Class2
Point x77 (value: 0.6317) is classified as Class2
Point x78 (value: 0.4718) is classified as Class1
Point x79 (value: 0.8997) is classified as Class2
Point x80 (value: 0.0340) is classified as Class1
Point x81 (value: 0.0145) is classified as Class1
Point x82 (value: 0.9525) is classified as Class2
Point x83 (value: 0.6636) is classified as Class2
Point x84 (value: 0.4974) is classified as Class2
Point x85 (value: 0.0442) is classified as Class1
Point x86 (value: 0.5753) is classified as Class2
Point x87 (value: 0.7679) is classified as Class2
Point x88 (value: 0.7419) is classified as Class2
Point x89 (value: 0.5881) is classified as Class2
Point x90 (value: 0.1566) is classified as Class1
Point x91 (value: 0.9331) is classified as Class2
Point x92 (value: 0.1519) is classified as Class1
Point x93 (value: 0.4980) is classified as Class2
Point x94 (value: 0.9087) is classified as Class2
Point x95 (value: 0.9070) is classified as Class2
Point x96 (value: 0.5238) is classified as Class2
Point x97 (value: 0.9585) is classified as Class2
Point x98 (value: 0.6557) is classified as Class2
Point x99 (value: 0.5350) is classified as Class2
Point x100 (value: 0.6439) is classified as Class2

Results for k = 2:

Point x51 (value: 0.7565) is classified as Class2
Point x52 (value: 0.0936) is classified as Class1
Point x53 (value: 0.9394) is classified as Class2
Point x54 (value: 0.0954) is classified as Class1
Point x55 (value: 0.0315) is classified as Class1
Point x56 (value: 0.0228) is classified as Class1
Point x57 (value: 0.3583) is classified as Class1
Point x58 (value: 0.6590) is classified as Class2
Point x59 (value: 0.6483) is classified as Class2
Point x60 (value: 0.4174) is classified as Class1
Point x61 (value: 0.1488) is classified as Class1
Point x62 (value: 0.4124) is classified as Class1
Point x63 (value: 0.4557) is classified as Class1
Point x64 (value: 0.2345) is classified as Class1
Point x65 (value: 0.9202) is classified as Class2
Point x66 (value: 0.1639) is classified as Class1
Point x67 (value: 0.2964) is classified as Class1
Point x68 (value: 0.1987) is classified as Class1
Point x69 (value: 0.6599) is classified as Class2
Point x70 (value: 0.7217) is classified as Class2
Point x71 (value: 0.5584) is classified as Class2
Point x72 (value: 0.8772) is classified as Class2
Point x73 (value: 0.7663) is classified as Class2
Point x74 (value: 0.1449) is classified as Class1
Point x75 (value: 0.8393) is classified as Class2
Point x76 (value: 0.9905) is classified as Class2
Point x77 (value: 0.6317) is classified as Class2
Point x78 (value: 0.4718) is classified as Class1
Point x79 (value: 0.8997) is classified as Class2
Point x80 (value: 0.0340) is classified as Class1
Point x81 (value: 0.0145) is classified as Class1
Point x82 (value: 0.9525) is classified as Class2
Point x83 (value: 0.6636) is classified as Class2
Point x84 (value: 0.4974) is classified as Class2
Point x85 (value: 0.0442) is classified as Class1
Point x86 (value: 0.5753) is classified as Class2
Point x87 (value: 0.7679) is classified as Class2
Point x88 (value: 0.7419) is classified as Class2
Point x89 (value: 0.5881) is classified as Class2
Point x90 (value: 0.1566) is classified as Class1
Point x91 (value: 0.9331) is classified as Class2
Point x92 (value: 0.1519) is classified as Class1
Point x93 (value: 0.4980) is classified as Class2
Point x94 (value: 0.9087) is classified as Class2
Point x95 (value: 0.9070) is classified as Class2
Point x96 (value: 0.5238) is classified as Class2
Point x97 (value: 0.9585) is classified as Class2
Point x98 (value: 0.6557) is classified as Class2
Point x99 (value: 0.5350) is classified as Class2
Point x100 (value: 0.6439) is classified as Class2

Results for k = 3:

Point x51 (value: 0.7565) is classified as Class2
Point x52 (value: 0.0936) is classified as Class1
Point x53 (value: 0.9394) is classified as Class2
Point x54 (value: 0.0954) is classified as Class1
Point x55 (value: 0.0315) is classified as Class1
Point x56 (value: 0.0228) is classified as Class1
Point x57 (value: 0.3583) is classified as Class1
Point x58 (value: 0.6590) is classified as Class2
Point x59 (value: 0.6483) is classified as Class2
Point x60 (value: 0.4174) is classified as Class1
Point x61 (value: 0.1488) is classified as Class1
Point x62 (value: 0.4124) is classified as Class1
Point x63 (value: 0.4557) is classified as Class1
Point x64 (value: 0.2345) is classified as Class1
Point x65 (value: 0.9202) is classified as Class2
Point x66 (value: 0.1639) is classified as Class1
Point x67 (value: 0.2964) is classified as Class1
Point x68 (value: 0.1987) is classified as Class1
Point x69 (value: 0.6599) is classified as Class2
Point x70 (value: 0.7217) is classified as Class2
Point x71 (value: 0.5584) is classified as Class2
Point x72 (value: 0.8772) is classified as Class2
Point x73 (value: 0.7663) is classified as Class2
Point x74 (value: 0.1449) is classified as Class1
Point x75 (value: 0.8393) is classified as Class2
Point x76 (value: 0.9905) is classified as Class2
Point x77 (value: 0.6317) is classified as Class2
Point x78 (value: 0.4718) is classified as Class1
Point x79 (value: 0.8997) is classified as Class2
Point x80 (value: 0.0340) is classified as Class1
Point x81 (value: 0.0145) is classified as Class1
Point x82 (value: 0.9525) is classified as Class2
Point x83 (value: 0.6636) is classified as Class2
Point x84 (value: 0.4974) is classified as Class2
Point x85 (value: 0.0442) is classified as Class1
Point x86 (value: 0.5753) is classified as Class2
Point x87 (value: 0.7679) is classified as Class2
Point x88 (value: 0.7419) is classified as Class2
Point x89 (value: 0.5881) is classified as Class2
Point x90 (value: 0.1566) is classified as Class1
Point x91 (value: 0.9331) is classified as Class2
Point x92 (value: 0.1519) is classified as Class1
Point x93 (value: 0.4980) is classified as Class2
Point x94 (value: 0.9087) is classified as Class2
Point x95 (value: 0.9070) is classified as Class2
Point x96 (value: 0.5238) is classified as Class2
Point x97 (value: 0.9585) is classified as Class2
Point x98 (value: 0.6557) is classified as Class2
Point x99 (value: 0.5350) is classified as Class2
Point x100 (value: 0.6439) is classified as Class2

Results for k = 4:
Point x51 (value: 0.7565) is classified as Class2
Point x52 (value: 0.0936) is classified as Class1
Point x53 (value: 0.9394) is classified as Class2
Point x54 (value: 0.0954) is classified as Class1
Point x55 (value: 0.0315) is classified as Class1
Point x56 (value: 0.0228) is classified as Class1
Point x57 (value: 0.3583) is classified as Class1
Point x58 (value: 0.6590) is classified as Class2
Point x59 (value: 0.6483) is classified as Class2
Point x60 (value: 0.4174) is classified as Class1
Point x61 (value: 0.1488) is classified as Class1
Point x62 (value: 0.4124) is classified as Class1
Point x63 (value: 0.4557) is classified as Class1
Point x64 (value: 0.2345) is classified as Class1
Point x65 (value: 0.9202) is classified as Class2
Point x66 (value: 0.1639) is classified as Class1
Point x67 (value: 0.2964) is classified as Class1
Point x68 (value: 0.1987) is classified as Class1
Point x69 (value: 0.6599) is classified as Class2
Point x70 (value: 0.7217) is classified as Class2
Point x71 (value: 0.5584) is classified as Class2
Point x72 (value: 0.8772) is classified as Class2
Point x73 (value: 0.7663) is classified as Class2
Point x74 (value: 0.1449) is classified as Class1
Point x75 (value: 0.8393) is classified as Class2
Point x76 (value: 0.9905) is classified as Class2
Point x77 (value: 0.6317) is classified as Class2
Point x78 (value: 0.4718) is classified as Class1
Point x79 (value: 0.8997) is classified as Class2
Point x80 (value: 0.0340) is classified as Class1
Point x81 (value: 0.0145) is classified as Class1
Point x82 (value: 0.9525) is classified as Class2
Point x83 (value: 0.6636) is classified as Class2
Point x84 (value: 0.4974) is classified as Class2
Point x85 (value: 0.0442) is classified as Class1
Point x86 (value: 0.5753) is classified as Class2
Point x87 (value: 0.7679) is classified as Class2
Point x88 (value: 0.7419) is classified as Class2
Point x89 (value: 0.5881) is classified as Class2
Point x90 (value: 0.1566) is classified as Class1
Point x91 (value: 0.9331) is classified as Class2
Point x92 (value: 0.1519) is classified as Class1
Point x93 (value: 0.4980) is classified as Class2
Point x94 (value: 0.9087) is classified as Class2
Point x95 (value: 0.9070) is classified as Class2
Point x96 (value: 0.5238) is classified as Class2
Point x97 (value: 0.9585) is classified as Class2
Point x98 (value: 0.6557) is classified as Class2
Point x99 (value: 0.5350) is classified as Class2
Point x100 (value: 0.6439) is classified as Class2

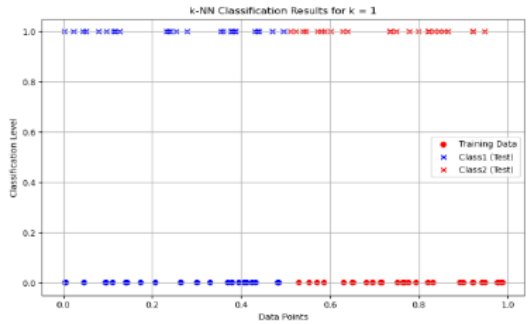
Results for k = 5:
Point x51 (value: 0.7565) is classified as Class2
Point x52 (value: 0.0936) is classified as Class1
Point x53 (value: 0.9394) is classified as Class2
Point x54 (value: 0.0954) is classified as Class1
Point x55 (value: 0.0315) is classified as Class1
Point x56 (value: 0.0228) is classified as Class1
Point x57 (value: 0.3583) is classified as Class1
Point x58 (value: 0.6590) is classified as Class2
Point x59 (value: 0.6483) is classified as Class2
Point x60 (value: 0.4174) is classified as Class1
Point x61 (value: 0.1488) is classified as Class1
Point x62 (value: 0.4124) is classified as Class1
Point x63 (value: 0.4557) is classified as Class1
Point x64 (value: 0.2345) is classified as Class1
Point x65 (value: 0.9202) is classified as Class2
Point x66 (value: 0.1639) is classified as Class1
Point x67 (value: 0.2964) is classified as Class1
Point x68 (value: 0.1987) is classified as Class1
Point x69 (value: 0.6599) is classified as Class2
Point x70 (value: 0.7217) is classified as Class2
Point x71 (value: 0.5584) is classified as Class2
Point x72 (value: 0.8772) is classified as Class2
Point x73 (value: 0.7663) is classified as Class2
Point x74 (value: 0.1449) is classified as Class1
Point x75 (value: 0.8393) is classified as Class2
Point x76 (value: 0.9905) is classified as Class2
Point x77 (value: 0.6317) is classified as Class2
Point x78 (value: 0.4718) is classified as Class1
Point x79 (value: 0.8997) is classified as Class2
Point x80 (value: 0.0340) is classified as Class1
Point x81 (value: 0.0145) is classified as Class1
Point x82 (value: 0.9525) is classified as Class2
Point x83 (value: 0.6636) is classified as Class2
Point x84 (value: 0.4974) is classified as Class1
Point x85 (value: 0.0442) is classified as Class1
Point x86 (value: 0.5753) is classified as Class2
Point x87 (value: 0.7679) is classified as Class2
Point x88 (value: 0.7419) is classified as Class2
Point x89 (value: 0.5881) is classified as Class2
Point x90 (value: 0.1566) is classified as Class1
Point x91 (value: 0.9331) is classified as Class2
Point x92 (value: 0.1519) is classified as Class1
Point x93 (value: 0.4980) is classified as Class1
Point x94 (value: 0.9087) is classified as Class2
Point x95 (value: 0.9070) is classified as Class2
Point x96 (value: 0.5238) is classified as Class2
Point x97 (value: 0.9585) is classified as Class2
Point x98 (value: 0.6557) is classified as Class2
Point x99 (value: 0.5350) is classified as Class2
Point x100 (value: 0.6439) is classified as Class2

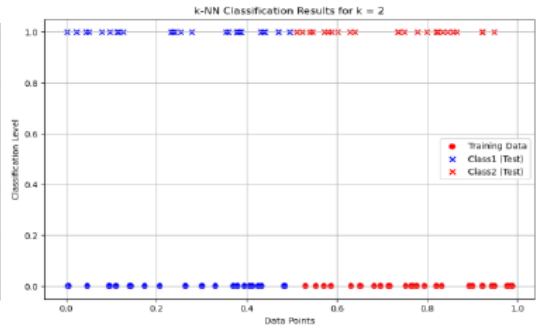
Results for k = 20:
Point x51 (value: 0.7565) is classified as Class2
Point x52 (value: 0.0936) is classified as Class1
Point x53 (value: 0.9394) is classified as Class2
Point x54 (value: 0.0954) is classified as Class1
Point x55 (value: 0.0315) is classified as Class1
Point x56 (value: 0.0228) is classified as Class1
Point x57 (value: 0.3583) is classified as Class1
Point x58 (value: 0.6590) is classified as Class2
Point x59 (value: 0.6483) is classified as Class2
Point x60 (value: 0.4174) is classified as Class1
Point x61 (value: 0.1488) is classified as Class1
Point x62 (value: 0.4124) is classified as Class1
Point x63 (value: 0.4557) is classified as Class1
Point x64 (value: 0.2345) is classified as Class1
Point x65 (value: 0.9202) is classified as Class2
Point x66 (value: 0.1639) is classified as Class1
Point x67 (value: 0.2964) is classified as Class1
Point x68 (value: 0.1987) is classified as Class1
Point x69 (value: 0.6599) is classified as Class2
Point x70 (value: 0.7217) is classified as Class2
Point x71 (value: 0.5584) is classified as Class2
Point x72 (value: 0.8772) is classified as Class2
Point x73 (value: 0.7663) is classified as Class2
Point x74 (value: 0.1449) is classified as Class1
Point x75 (value: 0.8393) is classified as Class2
Point x76 (value: 0.9905) is classified as Class2
Point x77 (value: 0.6317) is classified as Class2
Point x78 (value: 0.4718) is classified as Class1
Point x79 (value: 0.8997) is classified as Class2
Point x80 (value: 0.0340) is classified as Class1
Point x81 (value: 0.0145) is classified as Class1
Point x82 (value: 0.9525) is classified as Class2
Point x83 (value: 0.6636) is classified as Class2
Point x84 (value: 0.4974) is classified as Class1
Point x85 (value: 0.0442) is classified as Class1
Point x86 (value: 0.5753) is classified as Class2
Point x87 (value: 0.7679) is classified as Class2
Point x88 (value: 0.7419) is classified as Class2
Point x89 (value: 0.5881) is classified as Class2
Point x90 (value: 0.1566) is classified as Class1
Point x91 (value: 0.9331) is classified as Class2
Point x92 (value: 0.1519) is classified as Class1
Point x93 (value: 0.4980) is classified as Class1
Point x94 (value: 0.9087) is classified as Class2
Point x95 (value: 0.9070) is classified as Class2
Point x96 (value: 0.5238) is classified as Class2
Point x97 (value: 0.9585) is classified as Class2
Point x98 (value: 0.6557) is classified as Class2
Point x99 (value: 0.5350) is classified as Class2
Point x100 (value: 0.6439) is classified as Class2

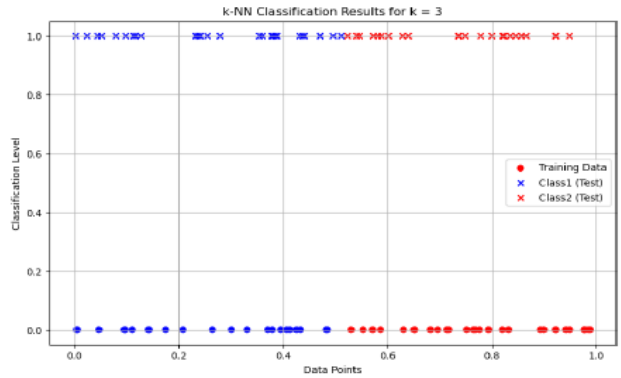
Results for k = 30:
Point x51 (value: 0.7565) is classified as Class2
Point x52 (value: 0.0936) is classified as Class1
Point x53 (value: 0.9394) is classified as Class2
Point x54 (value: 0.0954) is classified as Class1
Point x55 (value: 0.0315) is classified as Class1
Point x56 (value: 0.0228) is classified as Class1
Point x57 (value: 0.3583) is classified as Class1
Point x58 (value: 0.6590) is classified as Class2
Point x59 (value: 0.6483) is classified as Class2
Point x60 (value: 0.4174) is classified as Class1
Point x61 (value: 0.1488) is classified as Class1
Point x62 (value: 0.4124) is classified as Class1
Point x63 (value: 0.4557) is classified as Class1
Point x64 (value: 0.2345) is classified as Class1
Point x65 (value: 0.9202) is classified as Class2
Point x66 (value: 0.1639) is classified as Class1
Point x67 (value: 0.2964) is classified as Class1
Point x68 (value: 0.1987) is classified as Class1
Point x69 (value: 0.6599) is classified as Class2
Point x70 (value: 0.7217) is classified as Class2
Point x71 (value: 0.5584) is classified as Class2
Point x72 (value: 0.8772) is classified as Class2
Point x73 (value: 0.7663) is classified as Class2
Point x74 (value: 0.1449) is classified as Class1
Point x75 (value: 0.8393) is classified as Class2
Point x76 (value: 0.9905) is classified as Class2
Point x77 (value: 0.6317) is classified as Class2
Point x78 (value: 0.4718) is classified as Class1
Point x79 (value: 0.8997) is classified as Class2
Point x80 (value: 0.0340) is classified as Class1
Point x81 (value: 0.0145) is classified as Class1
Point x82 (value: 0.9525) is classified as Class2
Point x83 (value: 0.6636) is classified as Class2
Point x84 (value: 0.4974) is classified as Class1
Point x85 (value: 0.0442) is classified as Class1
Point x86 (value: 0.5753) is classified as Class2
Point x87 (value: 0.7679) is classified as Class2
Point x88 (value: 0.7419) is classified as Class2
Point x89 (value: 0.5881) is classified as Class2
Point x90 (value: 0.1566) is classified as Class1
Point x91 (value: 0.9331) is classified as Class2
Point x92 (value: 0.1519) is classified as Class1
Point x93 (value: 0.4980) is classified as Class1
Point x94 (value: 0.9087) is classified as Class2
Point x95 (value: 0.9070) is classified as Class2
Point x96 (value: 0.5238) is classified as Class2
Point x97 (value: 0.9585) is classified as Class2
Point x98 (value: 0.6557) is classified as Class2
Point x99 (value: 0.5350) is classified as Class2
Point x100 (value: 0.6439) is classified as Class2

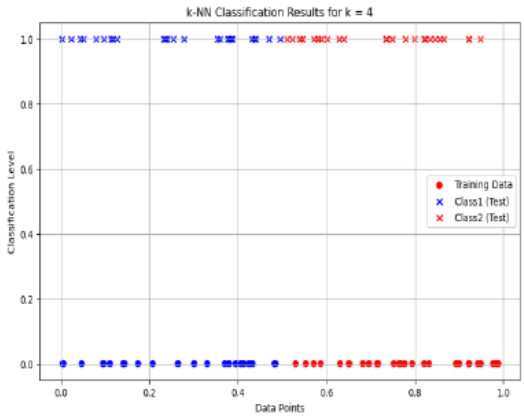
Classification complete.

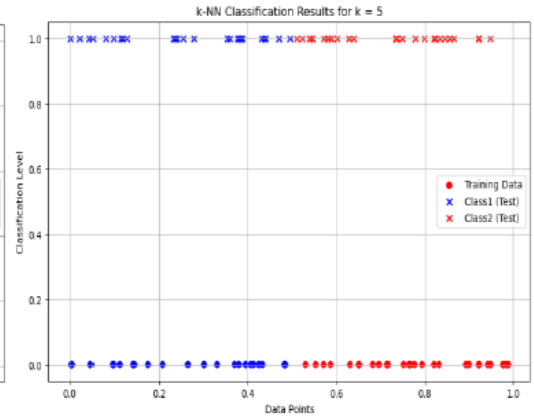
LAB 5

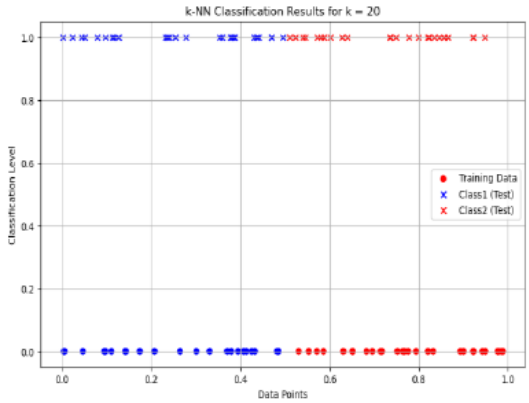

k-NN Classification Results for k = 1
Classification Level
Data Points
Training Data
Class1 (Test)
Class2 (Test)

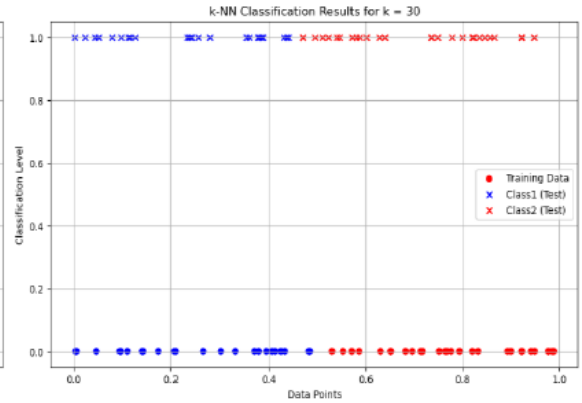

k-NN Classification Results for k = 2
Classification Level
Data Points
Training Data
Class1 (Test)
Class2 (Test)


k-NN Classification Results for k = 3
Classification Level
Data Points
Training Data
Class1 (Test)
Class2 (Test)

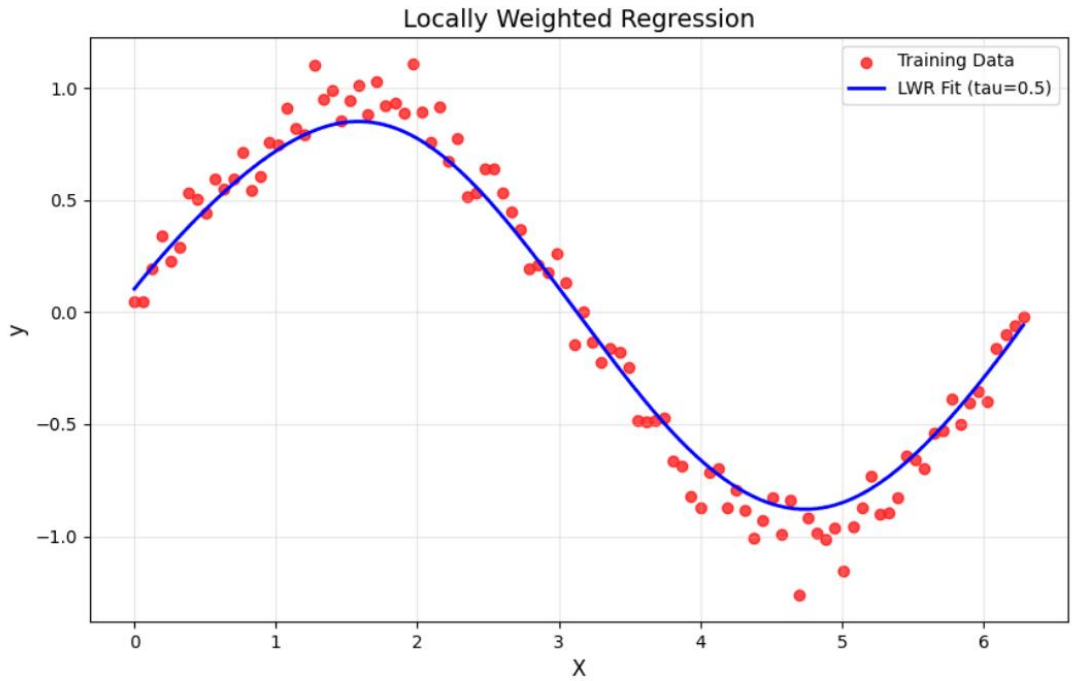

k-NN Classification Results for k = 4
Classification Level
Data Points
Training Data
Class1 (Test)
Class2 (Test)


k-NN Classification Results for k = 5
Classification Level
Data Points
Training Data
Class1 (Test)
Class2 (Test)

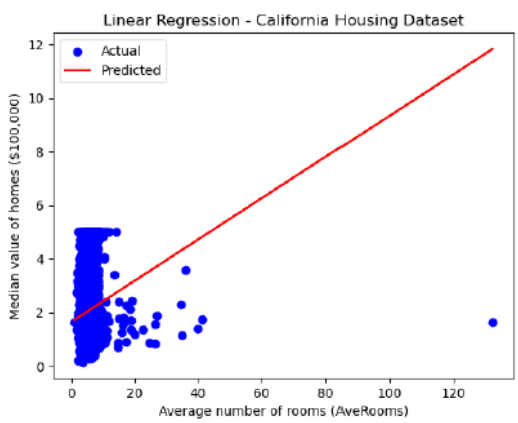

k-NN Classification Results for k = 20
Classification Level
Data Points
Training Data
Class1 (Test)
Class2 (Test)

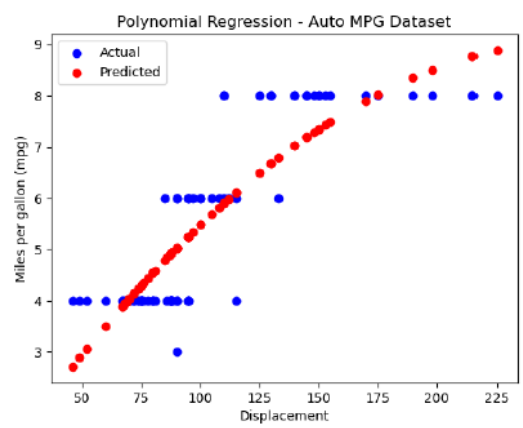

k-NN Classification Results for k = 30
Classification Level
Data Points
Training Data
Class1 (Test)
Class2 (Test)

LAB 6



LAB 7





Linear Regression - California Housing Dataset

Mean Squared Error: 1.2923314440807299

R^2 Score: 0.013795337532284901

Polynomial Regression - Auto MPG Dataset

Mean Squared Error: 0.7431490557205862

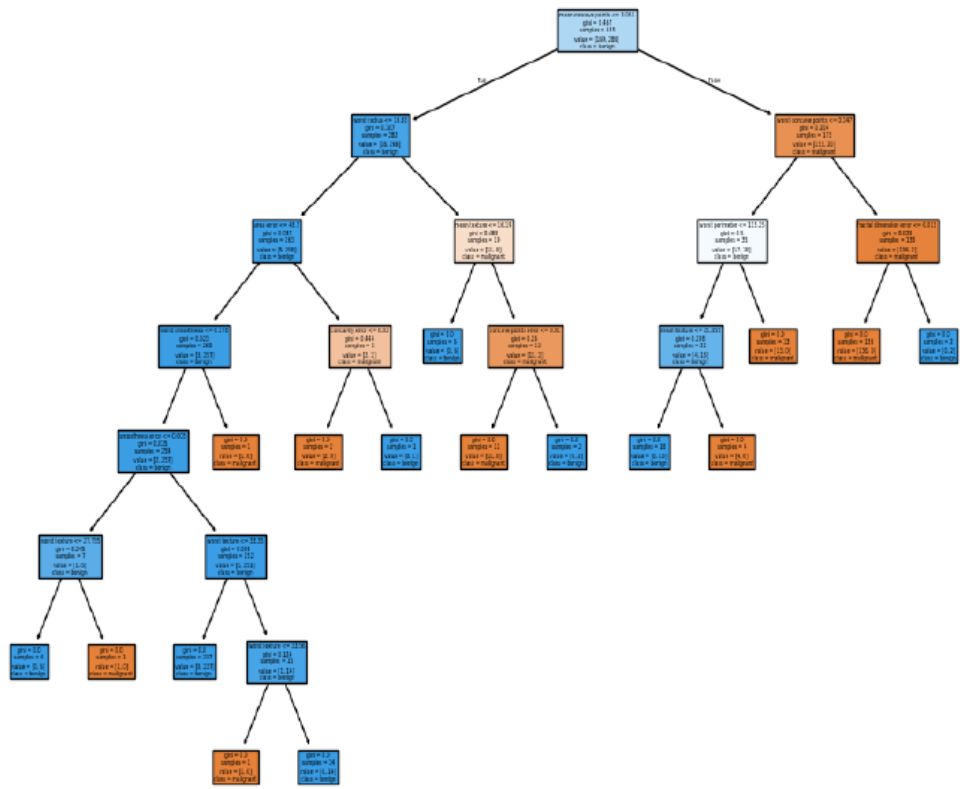
R^2 Score: 0.7505650609469626

LAB 8

Model Accuracy: 94.74%

Predicted Class for the new sample: Benign

Decision Tree - Breast Cancer Dataset



LAB 9

Accuracy: 80.83%

Classification Report:

	precision	recall	f1-score	support
0	0.67	1.00	0.80	2
1	1.00	1.00	1.00	2
2	0.33	0.67	0.44	3
3	1.00	0.00	0.00	5
4	1.00	0.50	0.67	4
5	1.00	1.00	1.00	2
7	1.00	0.75	0.86	4
8	1.00	0.67	0.80	3
9	1.00	0.75	0.86	4
10	1.00	1.00	1.00	3
11	1.00	1.00	1.00	1
12	0.40	1.00	0.57	4
13	1.00	0.80	0.89	5
14	1.00	0.40	0.57	5
15	0.67	1.00	0.80	2
16	1.00	0.67	0.80	3
17	1.00	1.00	1.00	3
18	1.00	1.00	1.00	3
19	0.67	1.00	0.80	2
20	1.00	1.00	1.00	3

...

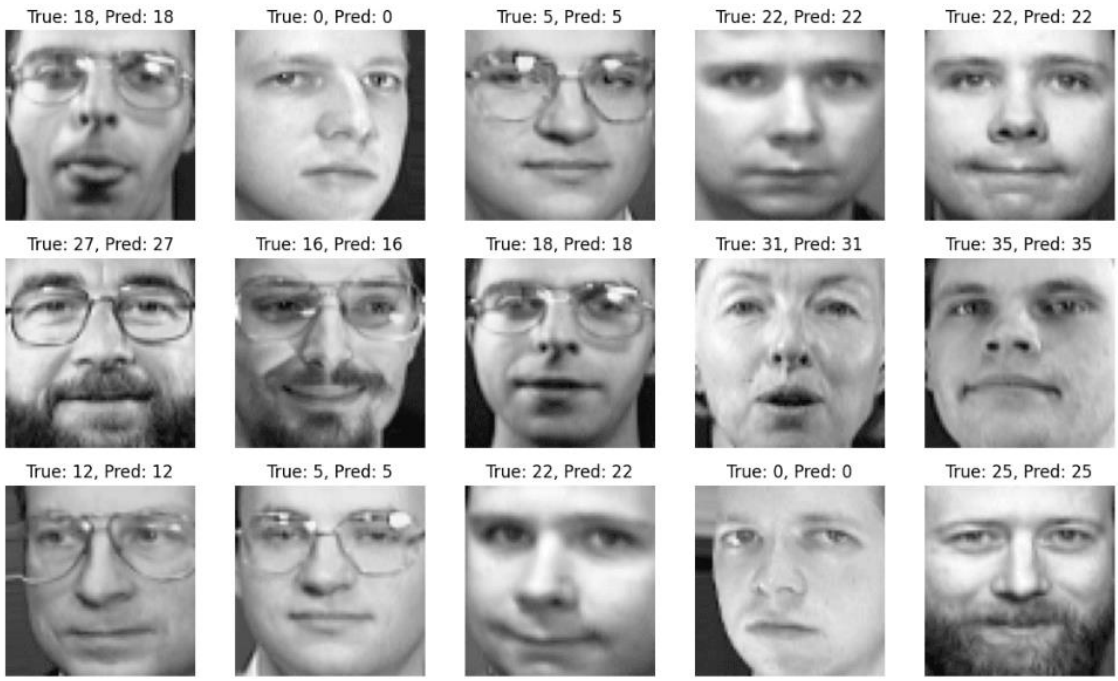
[0 0 0 ... 0 3 0]

[0 0 0 ... 0 0 5]]

Cross-validation accuracy: 87.25%

Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...

LAB 9

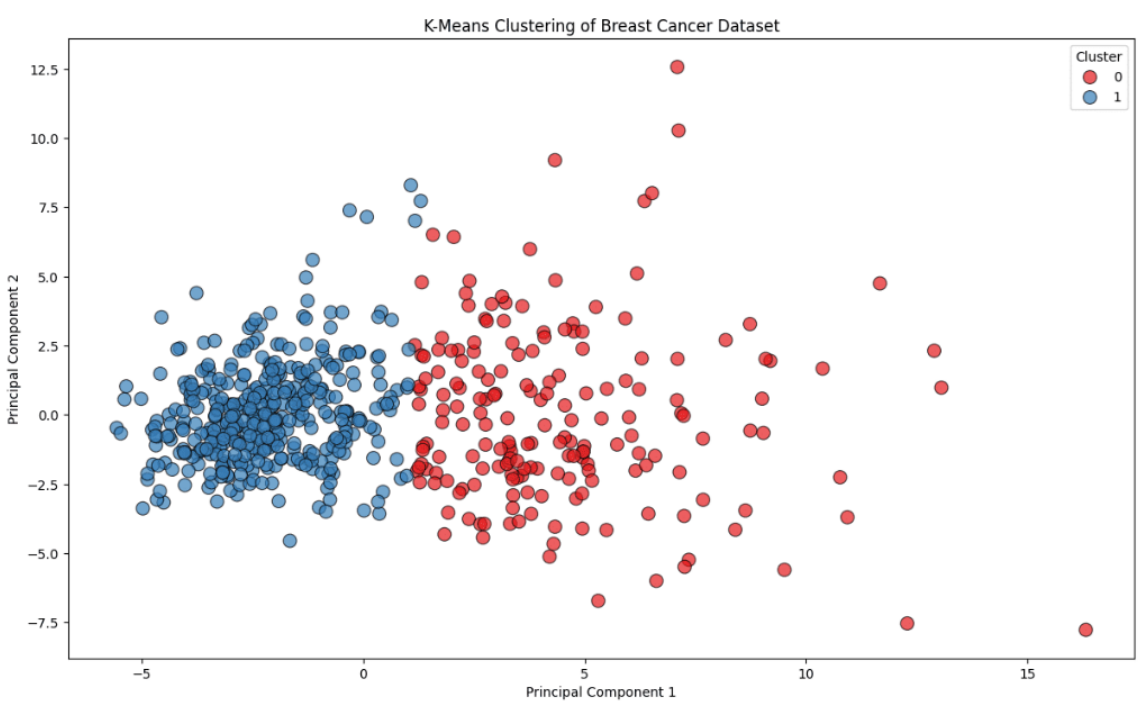


LAB 10

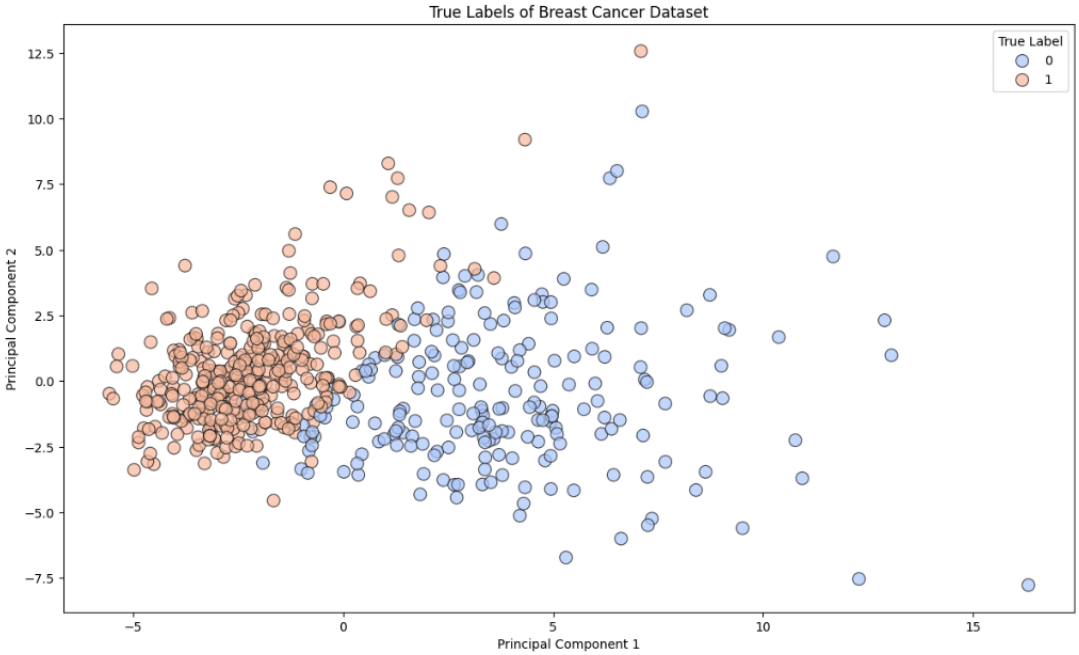
Confusion Matrix:
[[175 37]
[13 344]]

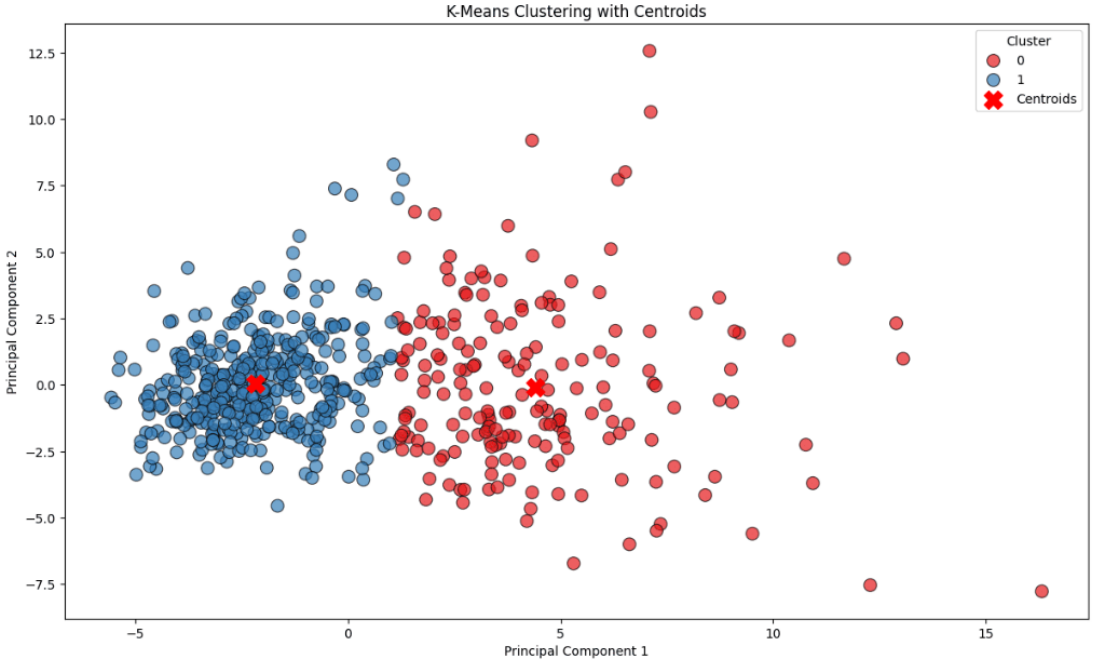
Classification Report:

	precision	recall	f1-score	support
0	0.93	0.83	0.88	212
1	0.90	0.96	0.93	357
accuracy			0.91	569
macro avg	0.92	0.89	0.90	569
weighted avg	0.91	0.91	0.91	569



LAB 10


True Labels of Breast Cancer Dataset
True Label
0
1
Principal Component 2
Principal Component 1
12.5
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−5
0
5
10
15


K-Means Clustering with Centroids
Cluster
0
1
Centroids
Principal Component 2
Principal Component 1
12.5
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−5
0
5
10
15